

JUNYEOP BAEK

Ph.D. Student @ KAIST

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RESEARCH INTEREST

My research goal is to build “*An embodied machine that can develop intelligence throughout its lifetime, just like humans.*” Specifically, the machine should establish and grow its own lens of worlds, goals, and values in open-ended environments — with interpretability and alignment as guiding concerns.

My current path toward this vision is to build **compositional world models** — understanding the world as reusable structure — and the **recursive reasoning** that runs over them to generalize beyond experience.

I pursue the following research questions, with **Q1** as my current focus throughout the PhD:

- **Q1. How can AI understand the world more systematically and compositionally? – Systematic World Understanding & Generalization (current focus)**
- *Q2. How can AI continuously accumulate and apply knowledge/skills? – Continual Learning*
- *Q3. Can AI develop its own goals following intrinsic motivation? – Autotelic Agents*
- *Q4. How can AI maintain autonomy while sharing human values? – Human-Value Alignment*

EDUCATION

Korea Advanced Institute of Science and Technology(KAIST)

Mar. 2025 – Present

Ph.D. in Computer Science - **Advisor:** Sungjin Ahn

Daejeon, Korea

Korea Advanced Institute of Science and Technology(KAIST)

Mar. 2023 – Feb. 2025

M.S. in Artificial Intelligence - **Advisor:** Sungjin Ahn

Daejeon, Korea

- **Thesis:** Dreamweaver: Learning Compositional World Models from Pixels

Hanyang University(ERICA)

Mar. 2014 – Feb. 2018

B.S. in Robot Engineering - **GPA** - 4.29 / 4.5

Ansan, Korea

- *Summa Cum Laude*
- *Capstone Project1: Disaster and Rescue Robot Simulation Modeling(Darrsm) - 2nd Prize in College*
- *Capstone Project2: Walk Yourself, Toddler!(WalkYTo) - 3rd Prize in Department*

RESEARCH EXPERIENCE

Visiting Researcher

Jun. 2025 – Aug. 2025

Mila-Quebec AI Institute

Montreal, Canada

- Collaborating with *Yoshua Bengio*’s research group and *LawZero* team members
- Conducting research project on **Mechanistic Interpretability and Chain-of-Thought Faithfulness in LLMs**
- **Main Collaborators:** Joumana Ghosn, Pierre-Luc St-Charles, Marc-Antoine Rondeau, Minsu Kim

HONORS & AWARDS

- **Best Paper Award**, ICML 2026 Workshop on Compositional Learning (2026)
- **Oral Presentation (Top 0.7%, 168/23918)**, ICML 2026
- **Silver Reviewer**, ICML 2026
- **Outstanding Teaching Assistant Award**, KAIST (2025)

PUBLICATIONS

1. **Junyeob Baek**^{*}, Mingyu Jo^{*}, Minsu Kim, Mengye Ren, Yoshua Bengio, and Sungjin Ahn. "Generative Recursive Reasoning". In: *Conference on Neural Information Processing Systems (NeurIPS)* (2026). (*Equal contribution)
2. Doojin Baek^{*}, **Junyeob Baek**^{*}, Mingyu Jo, Hosung Lee, Taegu Kang, Gyubin Lee, and Sungjin Ahn. "BabyTheorist: A Benchmark for Learning to Theorize the World from Observation Alone". In: *Conference on Neural Information Processing Systems (NeurIPS)* (2026). (*Equal contribution)
3. Doojin Baek^{*}, Gyubin Lee^{*}, **Junyeob Baek**, Hosung Lee, and Sungjin Ahn. "Learning to Theorize the World from Observation". In: *International Conference on Machine Learning (ICML)* (2026). **Oral Presentation (Top 0.7%) • Best Paper Award**, ICML 2026 Workshop on Compositional Learning. (*Equal contribution)
4. **Junyeob Baek**, Hosung Lee, Christopher Hoang, Mengye Ren, and Sungjin Ahn. "Discrete JEPA: Learning Discrete Token Representations without Reconstruction". In: *International Conference on Machine Learning (ICML) Tokenization Workshop* (2025).
5. Donghwan Chi, Hyomin Kim, Yoojin Oh, Yongjin Kim, Donghoon Lee, Daejin Jo, Jongmin Kim, **Junyeob Baek**, Sungjin Ahn, and Sungwoong Kim. "Slot-MLLM: Object-Centric Visual Tokenization for Multimodal LLM". *arXiv preprint arXiv:2505.17726* (2025).
6. **Junyeob Baek**, Yi-Fu Wu, Gautam Singh, and Sungjin Ahn. "Dreamweaver: Learning Compositional World Models from Pixels". In: *International Conference on Learning Representation (ICLR)* (2025).
7. Chang Chen, **Junyeob Baek**, Fei Deng, Kenji Kawaguchi, Caglar Gulcehre, and Sungjin Ahn. "PlanDQ: Hierarchical Plan Orchestration via D-Conductor and Q-Performer". In: *International Conference on Machine Learning (ICML)* (2024).

THESIS

- **Junyeob Baek**. "Dreamweaver: Learning Compositional World Models from Pixels". Master's Thesis, KAIST, Advisor: Sungjin Ahn (Feb. 2025).

ACADEMIC SERVICE

Conference/Journal Reviewer

- International Conference on Learning Representations (ICLR) 2025–2027
- International Conference on Machine Learning (ICML) 2026 — **Silver Reviewer**
- Conference on Neural Information Processing Systems (NeurIPS) 2026
- Transactions on Machine Learning Research (TMLR) 2026–

Workshop Organizer

- 3rd KAIST-Mila Prefrontal AI Workshop (6 invited speakers incl. Yoshua Bengio), Mila, Aug. 2026, [Link](#)
- 2nd KAIST-Mila Prefrontal AI Workshop (6 invited speakers incl. Yoshua Bengio), Mila, Jul. 2025, [Link](#)

TEACHING EXPERIENCE

Teaching Assistant

CS30707: Introduction to Reinforcement Learning, KAIST

Spring 2026

Teaching Assistant

CS672: Reinforcement Learning, KAIST

Fall 2025

Teaching Assistant

CS377: Introduction to Reinforcement Learning, KAIST

Spring 2025

- Outstanding Teaching Assistant Award recipient

PROFESSIONAL EXPERIENCE

MakinaRocks

Jun. 2021 – Jun. 2022

ML Research Engineer(Robotics Focused)

Seoul, Korea

- **Optimization of Welding Task and Motion Planning in Vehicle Manufacturing Process**
 - * Challenges: 1. Large-scale and high complexity in Optimization Problem. 2. Required high-accuracy planning in highly cluttered environments. 3. Required tight Time Window for control
 - * [*Contribution 1*] GA-based Multi-objective Optimization Module for Welding-point Assignment
 - * [*Contribution 2*] Multi-Robot Interlock Scheduling System with Swept-volume-based collision detection

Syscon, Co., Ltd.

Dec. 2017 – Jun. 2021

Robotics Software Engineer/Research Strategy Manager

Incheon, Korea

- **AMR Navigation System Design & Product Development**
- [*Field Exp.*] **Successful Deployment for Smart Factory Project**
 - * More than 300 delivery tasks per day with 14 robots
- [*Research Exp.*] **Learning-based High-accuracy Object Pose Estimator(Dock Pose Estimation)**
- [*Research Exp.*] **Distilled Meta Policy Navigation System for Multi-behavior Skillset and its Strategies**